

OMAHA MILLARD, NE, USA

WMO: 720308

Lat: 41.196N

Lon: 96.112W

Elev: 1049

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	4.5	-9.0	3.5	2.6	-5.7	4.2	5.8	25.8	27.4	23.5	25.9	5.4	320	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	94.8	75.3	91.3	74.7	89.1	73.6	79.1	89.4	77.3	87.5	75.7	85.6	9.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	142.6	86.6	74.1	132.2	83.9	72.6	125.4	82.4	43.4	89.2	41.4	87.9	39.8	85.6	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.7	16.6	-6.3	99.2	3.8	2.5	-9.1	101.1	-11.3	102.5	-13.5	103.9	-16.3	105.8		
(5)			-7.1	82.0	3.7	1.5	-9.7	83.1	-11.9	84.0	-13.9	84.8	-16.6	85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.8	25.8	28.3	42.3	52.8	63.7	73.7	77.5	74.7	68.0	54.0	41.5	29.4		
	DBStd	20.43	11.41	11.52	11.81	9.79	8.79	6.10	5.86	5.47	7.82	9.17	9.60	10.50		
	HDD50	2734	751	610	299	84	7	0	0	0	0	60	283	640		
	HDD65	5701	1216	1027	708	380	134	6	1	4	57	358	706	1104		
	CDD50	3744	0	3	59	167	430	711	852	767	541	186	27	1		
	CDD65	1234	0	0	4	13	92	266	388	305	148	18	0	0		
	CDH74	12225	0	0	51	177	929	2657	4111	2742	1361	191	6	0		
	CDH80	4676	0	0	10	44	302	1027	1768	1008	468	49	0	0		
	WSAvg	6.3	6.4	6.5	7.3	7.7	6.7	6.2	5.1	4.9	5.8	6.2	6.3	6.3		
	PrecAvg	31.3	0.9	1.0	1.9	3.0	4.3	5.0	4.1	3.9	2.6	2.2	1.5	0.9		
(15) Precipitation	PrecMax	43.5	1.5	2.8	3.5	6.2	9.3	10.6	11.4	8.6	5.7	5.4	2.6	2.6		
	PrecMin	24.9	0.3	0.2	0.1	0.4	1.4	1.4	1.3	1.1	0.6	0.0	0.0	0.0		
	PrecStd	5.2	0.4	0.6	1.1	1.3	2.2	2.6	2.5	2.2	1.3	1.5	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	67.2	79.8	86.1	93.6	96.2	99.4	96.0	93.4	86.4	73.3	58.7		
		MCWB	44.6	51.9	60.5	63.4	69.6	75.2	75.0	75.0	74.6	69.5	56.3	48.0		
	2%	DB	51.0	57.1	73.1	79.1	87.2	92.0	95.4	91.7	89.8	79.0	67.4	53.3		
		MCWB	42.3	46.7	57.0	60.9	67.3	74.7	76.3	75.1	73.1	63.8	54.6	44.9		
	5%	DB	45.6	50.4	67.5	73.5	82.5	89.4	91.8	88.7	86.0	74.2	61.5	48.3		
		MCWB	39.2	42.5	54.8	58.0	65.6	73.5	76.0	74.6	71.3	60.3	50.9	41.6		
	10%	DB	41.4	44.7	61.3	69.5	79.3	86.2	88.9	85.9	81.7	70.0	56.8	43.8		
		MCWB	36.9	39.1	51.9	56.2	64.1	71.5	74.8	73.4	68.5	58.0	47.8	38.5		
	0.4%	WB	46.1	53.6	63.4	66.4	73.3	80.2	82.2	80.3	77.7	71.3	60.2	52.4		
		MCDB	55.2	66.3	74.4	79.1	86.1	90.2	93.7	90.6	89.2	81.6	67.3	55.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.9	47.1	60.4	63.2	71.1	77.5	79.8	78.0	75.6	67.3	56.1	45.8		
		MCDB	49.7	57.1	68.5	74.6	83.6	89.0	90.4	87.4	85.2	75.8	64.9	50.7		
	5%	WB	40.0	42.8	56.3	60.6	68.7	75.5	77.6	76.3	73.8	63.6	52.6	42.1		
		MCDB	45.6	49.6	65.7	70.8	79.1	86.0	87.6	85.6	81.9	71.2	60.8	48.2		
	10%	WB	36.8	39.2	51.7	58.2	66.4	73.4	75.9	74.6	71.3	59.4	48.4	39.0		
		MCDB	41.5	44.8	63.4	67.1	76.2	83.4	85.9	83.7	79.0	67.2	56.3	43.7		
	5% DB	MDBR	17.0	17.5	20.0	20.7	20.3	19.1	18.3	18.3	20.8	20.8	19.7	16.0		
		MCDBR	24.4	27.5	30.3	28.6	25.5	21.8	21.4	21.9	24.3	27.3	28.3	24.1		
		MCWBR	16.9	17.7	16.6	14.7	10.7	9.5	8.6	9.5	10.3	14.1	16.4	17.0		
		MCWBR	22.7	25.6	24.0	23.8	21.9	19.3	18.8	18.7	19.7	22.7	25.7	22.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.0	17.3	14.1	13.9	10.4	10.2	9.5	9.7	9.8	14.2	16.6	16.4		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.318	0.332	0.393	0.417	0.422	0.431	0.421	0.380	0.333	0.300	0.285		
	taud		2.455	2.370	2.393	2.238	2.241	2.287	2.257	2.289	2.359	2.478	2.499	2.456		
	Ebn at Noon		280	283	291	279	273	271	267	267	271	274	271	268		
	Edh at Noon		24	31	34	43	43	42	43	40	35	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		642	938	1217	1555	1842	2050	2063	1783	1470	1031	720	541		
	RadStd		57	56	109	108	140	156	134	113	95	119	50	53		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+3.13	+3.55	N/A	N/A	N/A	N/A	
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page